ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TASK 2

Feature Engineering, Model Optimization and Performance Comparison

California Housing Price Prediction

Prepared using Python, Pandas, NumPy, Matplotlib and Scikit-learn

1. Introduction

This project applies machine learning techniques to the California Housing dataset to predict median house values. It explores data preprocessing, feature scaling, feature engineering, model training, hyperparameter optimization and performance evaluation.

2. Objectives

• Explore the dataset and identify missing values.

• Separate input features and the target variable.

• Split the dataset into training and testing sets.

• Apply feature scaling where appropriate.

• Train and compare three regression algorithms.

• Optimize Ridge Regression using different alpha values.

• Create prediction visualizations and examine feature importance.

• Evaluate engineered features using regression metrics.

3. Dataset Description

The California Housing dataset is loaded through Scikit-learn. It contains 20,640 observations and eight input features. The target variable is MedHouseVal, representing the median house value in the dataset's specified units.

Item	Description
Dataset size	20,640 rows
Input features	8 original features
Target variable	MedHouseVal
Missing values	None reported in the initial inspection

4. Data Preprocessing

The MedHouseVal column was separated as the target variable, while the remaining columns were used as input features. The data was split into 80% training data and 20% testing data using random_state=42. StandardScaler was fitted on the training features and then applied to both training and testing features to prevent information leakage.

5. Machine Learning Models

Model	Purpose
Linear Regression	Provides a baseline regression model.
Ridge Regression	Uses regularization to limit large coefficients.
Decision Tree Regressor	Learns nonlinear patterns using tree-based splits.

6. Evaluation Metrics

• Mean Absolute Error (MAE): Measures the average absolute prediction error.

• Mean Squared Error (MSE): Measures the average squared prediction error.

• Root Mean Squared Error (RMSE): Expresses the square root of MSE.

• R-squared (R^2): Measures how much variation in the target is explained by the model.

7. Model Optimization

Ridge Regression was evaluated with alpha values of 0.01, 0.1, 1, 10 and 100. The value producing the lowest RMSE among those tested was selected for the optimized Ridge model. This is a comparison of the tested values, not a guarantee of a globally optimal value.

8. Feature Engineering

Three additional features were created: RoomsPerHousehold, BedroomsPerRoom and PopulationPerHousehold. Linear Regression was trained using the original features and then using the engineered features. MAE, RMSE and R^2 were compared to assess the effect of the additional variables.

9. Visual Analysis

An Actual vs. Predicted scatter plot was created for the optimized Ridge Regression model. A feature-importance plot was also generated from the Decision Tree Regressor to examine which input features contributed most to its predictions.

10. Conclusion

This project demonstrates a complete regression workflow, including dataset exploration, preprocessing, feature scaling, model training, hyperparameter comparison, feature engineering and performance evaluation. The final model should be selected based on the observed metrics. The numerical comparison results can be added to this report after reviewing the completed notebook.